SYSTEM DESIGN SPECIFICATION — SOFTWARE ENGINEERING REPOSITORY

Complete Project Data Dictionary (DD)

Comprehensive Metadata Repository for Data Stores, Composite Structures, Field Elements & DFD Flows

Project Name:	CareerForge AI — Autonomous Career Readiness Platform
SE Standard:	IEEE Std 830-1998 (SRS) & Gane-Sarson / Yourdon Structured Analysis Standards
Document Type:	Project Data Dictionary (Companion to Level 0, Level 1 & Level 2 DFDs)
Target Database:	Supabase Cloud PostgreSQL (JSONB Architecture) + Client Local Storage Cache
Scope Covered:	All 4 Data Stores (D1–D4), 38 Individual Data Elements, 7 Composite Schemas, and 12 DFD Flows

Definition of Data Dictionary: In Software Engineering, a *Data Dictionary* is a centralized, authoritative catalog that describes every data element, data store, and data flow occurring in the system. It eliminates ambiguity between front-end developers, database administrators, and external evaluators by defining exact data types, lengths, ranges, nullability, validation rules, and structural compositions.

1. Data Stores Dictionary (The D1 – D4 Repositories)

The system organizes state across cloud relational stores and in-memory static catalogs:

ID	Data Store Name	Physical Location	Primary Key / Index	Structure / Schema Description	Access Role
D1	**User Profile Data** (auth.users)	Supabase Auth (PostgreSQL)	id (UUID PK) email (Unique Index)	Relational table storing authentication identities, bcrypt password hashes, email confirmation flags, and JWT session metadata.	System Auth Engine (Internal Read/Write)
D2	**Assessment & State Data** (public.student_states)	Supabase Public (PostgreSQL)	id (UUID PK) user_id (Foreign Key Index)	Hybrid relational table containing a flexible state_data (JSONB) column that persists full student assessment, resume, interview, and telemetry objects.	Student (RLS CRUD) Faculty (Read / Audit)
D3	**Career Domain Catalog** (DOMAINS)	In-Memory Client / Codebase	id (String Slug, e.g. 'frontend', 'aiml')	Static array of 25 domain objects containing required skills, industry tools, potential roles, RIASEC archetypes, and curated fallback question banks.	Read-Only (Public Reference)
D4	**Progress & Reward Data** (cert_registry)	Embedded in D2 / Faculty Ledger	verification_hash (SHA-256)	Digital ledger storing certificate status ('Not Eligible', 'Pending', 'Approved'), university endorsement signatures, badge collections, and gamified points.	Faculty (Write/Sign) Public (Verification)

2. Individual Data Elements Dictionary

Every individual variable, column, and property processed in CareerForge AI:

Field Name	Entity / Parent	Type	Length / Constraints	Description	Sample Value
id	auth.users	UUID	36 chars, PK, Not Null	Unique Supabase identity identifier	a7e82b...-91b7
email	Account	VARCHAR	Max 100, Valid Email format	Primary login email address	girish@uni.edu
name	Account	VARCHAR	Max 80, Alphabetical & spaces	Full candidate display name	Girish Patil
enrollment	Profile	VARCHAR	10–14 chars, Alphanumeric	University roll or enrollment number	2201010045

dept	Profile	VARCHAR	Enum: CS, IT, AI, EC, ME	Academic engineering department	Computer Science
year	Profile	TINYINT	Range: 1 to 4	Current year of study in B.Tech/BE	4
university	Profile	VARCHAR	Max 120 chars	Registered university or college name	State Tech University
domain	student_states	VARCHAR	Foreign Key to D3 id (Max 30)	Current selected career pathway slug	'aiml'
answers	Assessment	ARRAY[INT]	12 integers, each 0 to 3	Candidate RIASEC response choices	[2, 3, 1, 0, ...]
assessment.score	Assessment	INT	0 to 100 percentage	Initial baseline career discovery score	85
fit_percentage	Recommendation	FLOAT	0.0 to 100.0%	Calculated compatibility with target domain	94.5%
resume.current	Resume	INT	Range: 0 to 100	Latest ATS scan match percentage	78
resume.best	Resume	INT	Range: 0 to 100, Monotonic	All-time highest ATS scan score	88
resume.attempts	Resume	INT	>= 0, Counter	Number of resume versions audited	4
missing_keywords	Resume	ARRAY[TEXT]	Up to 10 strings	ATS identified gaps in technical stack	['Docker', 'Redis']
tech.best	Tech Interview	INT	Range: 0 to 100	Peak score in domain technical MCQ drills	90
hr.best	HR Interview	INT	Range: 0 to 100	Peak score in behavioral situational drills	85
mock.best	Mock Session	INT	Range: 0 to 100	Peak score in full AI voice/written mock	82
vowel_ratio	Anti-Gibberish	FLOAT	0.00 to 1.00 (Threshold: 0.12)	Ratio of vowels to total letters in answer	0.38 (Valid)
eval_relevance	LLM Evaluation	INT	Range: 0 to 25	Rubric sub-score: Query direct relevance	22
eval_technical	LLM Evaluation	INT	Range: 0 to 25	Rubric sub-score: Technical accuracy & depth	21
eval_comm	LLM Evaluation	INT	Range: 0 to 25	Rubric sub-score: Articulation & tone	20
eval_structure	LLM Evaluation	INT	Range: 0 to 25	Rubric sub-score: STAR format structure	19
readiness	Scorecard	INT	Range: 0 to 100%	Composite weighted institutional readiness	86
points	Rewards	INT	>= 0, Cumulative	Gamified XP points for completed drills	1250
certStatus	Certification	ENUM	'Not Eligible' | 'Pending' | 'Approved'	Faculty validation workflow status	'Approved'
certHash	Certification	CHAR(64)	Hex SHA-256	Cryptographic tamper-proof seal	7f4a9b...c1

3. Composite Data Structures (Data Records)

In modern web applications, high-level entities are bundled into composite structures. Below are the definitive composite schemas for CareerForge AI:

Structure 1: Student Master State Object (JSONB schema in D2)

```
{
  "account":    { "name": String, "email": String },
  "profile":    { "enrollment": String, "dept": String, "year": Number, "university": String, "skills": [String], "interests":
 [String] },
  "domain":     String,                      // Domain ID slug e.g. "aiml"
  "assessment": { "answers": [Number], "score": Number },
  "resume":     { "best": Number, "current": Number, "attempts": Number },
  "hr":         { "best": Number, "current": Number, "attempts": Number },
  "tech":       { "best": Number, "current": Number, "attempts": Number },
  "mock":       { "best": Number, "current": Number, "attempts": Number },
  "points":     Number,
  "pointHistory": [ { "date": ISOString, "desc": String, "pts": Number } ],
  "certStatus": "Not Eligible" | "Pending Faculty Review" | "Approved"
}
```

Structure 2: Career Domain Catalog Record (D3 Definition)

```
{
  "id":     String,                        // e.g. "aiml"
  "name":   String,                        // e.g. "Artificial Intelligence & ML"
  "cat":    String,                        // "Engineering" | "Data" | "Design" | "Infrastructure"
  "icon":   String,                        // Emoji icon e.g. "■"
  "desc":   String,                        // 1-sentence domain mission
  "skills": [ String ],                    // Core required technical skills (e.g. ["Python", "PyTorch"])
  "tools":  [ String ],                    // Standard industry tools (e.g. ["Jupyter", "Docker"])
  "roles":  [ String ],                    // Target entry-level jobs (e.g. ["ML Engineer", "Data Scientist"])
  "fit":    String                         // Psychological / aptitude fit statement
}
```

Structure 3: Gemini Evaluation Response Payload (External Flow)

```
{
  "relevance":     Number,                 // 0 to 25: Relevance of candidate response to question
  "technical":     Number,                 // 0 to 25: Depth of algorithmic / domain knowledge
  "communication": Number,                 // 0 to 25: Articulation, clarity, vocabulary
  "structure":     Number,                 // 0 to 25: Adherence to STAR behavioral structure
  "total":         Number,                 // 0 to 100: Sum of sub-scores
  "feedback":      String,                 // Detailed feedback highlighting strengths
  "star_tip":      String                  // Actionable improvement recommendation
}
```

Structure 4: Faculty Cohort Audit Dossier (Admin View Output)

```
{
  "id":          UUID,                     // Supabase Auth User ID
  "enrollment":  String,                   // College Enrollment Number
  "name":        String,                   // Candidate Name
  "email":       String,                   // Institutional Email
  "dept":        String,                   // Branch (e.g. "Computer Science")
  "domain":      String,                   // Chosen Career Pathway
  "readiness":   Number,                   // Composite Readiness Percentage (0-100)
  "ats":         Number,                   // Peak Resume Score
  "tech":        Number,                   // Peak Technical Interview Score
  "hr":          Number,                   // Peak HR Interview Score
  "mock":        Number,                   // Peak Mock Interview Score
  "cert":        String                    // Certification Approval Status
}
```

4. Data Flows Dictionary (DFD Mapping)

Every named arrow appearing in DFD Level 0, Level 1, and Level 2, along with its source, destination, and payload:

Data Flow Name	Source	Destination	Data Structure Carried	Purpose / Trigger
User Credentials	Student	Proc 1.0 (Auth)	{ email, password, name }	Login or signup request handshake.
Profile & Resume Draft	Student	Proc 2.0 / Proc 3.0	{ enrollment, dept, raw_text }	Submission of profile details or resume text for ATS audit.
Assessment Answers	Student	Proc 3.0 (Assessment)	answers[0..11] (Integers)	12 personality/aptitude responses submitted to discovery engine.
Career Recommendations	Proc 4.0	Student	Top 5 Domain Objects + Fit %	Delivers ranked domains, skills, tools, and 16:9 visual roadmap.
Written / Voice Response	Student	Proc 6.0 (Mock)	{ transcript, duration, domain }	Candidate interview answer submitted for anti-gibberish & LLM grading.
Gemini System Prompt	CareerForge Engine	Google Gemini API	REST JSON { role, question, answer, rubric }	HTTPS POST request asking LLM to generate or evaluate questions.
AI Structured Evaluation	Google Gemini API	CareerForge Engine	JSON { relevance, tech, comm, struct, tip }	HTTPS Response returning multi-dimensional score matrix.
State Upsert Mutation	All Processes	D2: student_states	JSONB state_data snapshot	Persists latest scores, attempts, and progress to Supabase Cloud.
Cohort Audit Query	Faculty / Admin	Proc 7.0 (Dashboard)	{ deptFilter, readinessRange, searchKey }	Administrator requests live placement analytics & student list.
Digital Certificate Endorsement	Faculty Admin	D4: cert_registry	{ hash, endorsedBy, timestamp, status }	Cryptographically validates student readiness for placement.

5. Data Transformation & Validation Rules

Rule Name	Applied In	Mathematical / Logical Rule	Outcome / Error Action
Anti-Gibberish Pre-Filter	Proc 6.0 (Mock Interview)	vowel_count / total_letters < 0.12 OR consecutive_consonants > 7	Flagged as keyboard mash. Bypasses Gemini API. Returns 0% score immediately with 'Invalid submission' warning.
ATS Keyword Match Score	Proc 3.0 (Resume)	$$\text{ATS} = \min\left(100, \frac{|\text{Found Skills}|}{|\text{Domain Skills}|} \times 70 + \text{LenBonus}\right)$$	Scores resume from 0 to 100%. Appends missing keywords to action checklist.
Composite Readiness Score	Proc 7.0 (Dashboard & Admin)	$$\text{Readiness} = \text{round}\left(\frac{\text{tech.best} + \text{hr.best} + \text{mock.best}}{N_{\text{active}}}\right)$$	Computes overall readiness percentage. Requires >= 75% for placement certification eligibility.
Certificate Endorsement	D4 / Admin Engine	Readiness >= 75% AND ATS >= 70% AND FacultyApproval == true	Generates 64-character SHA-256 digital stamp and unlocks printable PDF certificate.

6. Top 5 Data Dictionary Viva Questions & Answers

Q1: What is a Data Dictionary, and why is it essential for CareerForge AI?

Spoken Answer: A Data Dictionary is the central repository containing formal definitions of every data element, data structure, data store, and data flow in the project. It ensures that our frontend form inputs, database columns in Supabase, and AI payloads in Gemini follow consistent types, ranges, and validation constraints.

Q2: How does your Data Dictionary support your Data Flow Diagrams (DFDs)?

Spoken Answer: A DFD shows *where* data flows graphically, but it cannot show data types, nullability, or internal structures. The Data Dictionary provides the textual specification for every labeled arrow (data flow) and cylinder/open-box (data store) shown in DFD Levels 0, 1, and 2.

Q3: What is the difference between a Data Element and a Composite Data Structure?

Spoken Answer: A Data Element is an atomic, indivisible field like `user_id`, `readiness_score`, or `email`. A Composite Data Structure is a collection of elements grouped together, such as the `student_states` JSONB document or the Gemini evaluation response payload.

Q4: Why does CareerForge AI use a JSONB column (`state_data`) in Data Store D2?

Spoken Answer: We use JSONB in Supabase PostgreSQL because a student's career preparation involves evolving, multi-tiered data—such as dynamic quiz answers, point history arrays, and interview attempt metrics. JSONB offers the schema flexibility of NoSQL while maintaining PostgreSQL's ACID compliance and indexing capabilities.

Q5: How does the Data Dictionary define data security and validation in your project?

Spoken Answer: It defines security at two tiers: first, authentication credentials in Store D1 are secured using bcrypt hashes and Supabase Row Level Security (RLS). Second, candidate inputs undergo client-side regex constraints (like the anti-gibberish vowel ratio filter) before reaching external AI APIs.